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ITCC508 Lab Exercise PT-F1 (Optimizing Chunk Size and Overlap): Initial Preps - Code Review, Test Questions, and Prompt Settings

Objective: To conduct a controlled experiment to determine which chunk size and overlap setting produces the best retrieval performance for a given document dataset using the provided FAISS Vector Store and Retrieval notebook.

Learning Outcomes:

- Create and compare multiple chunk-size and overlap configurations.
- Evaluate retrieval with Precision@3, Recall@3, and MRR@3.
- Compare results for easy, average, and difficult questions.
- Recommend a dataset-specific configuration using measured evidence and efficiency.

Step 1: Code Commenting

Split the code in each notebook cell into individual cells based on logical functional sections. For every code cell, insert a Markdown Text Cell directly above it. Use the Text Cells to provide a step-by-step technical explanation detailing the specific function of each section.

Make sure to preserve the notebook's original functionality and verify that it executes correctly before making moving on with the next steps.

Step 2: Select your dataset

Choose a pdf document appropriate for your group's RAG application. Before proceeding, ensure that:

1. The document contain extractable text.

2. Run the document through the Vector Indexing and Embedding extraction and verify that you are able to generate a valid FAISS vector store by inspecting the chunks in the generated xlsx file.

Use the same dataset throughout the experiment.

Step 2: Prepare the test questions

Create at least three questions based on your own dataset and determine the correct answer and supporting evidence before running the experiments. The Table below will guide you in generating appropriate questions.

Level	Minimum requirement	Example (adapt to your dataset)	Expected Answer	Source Evidence
Easy	Single explicit fact in one passage	What is the official short title of Republic Act No. 9710?	The official short title of Republic Act No. 9710 is "The Magna Carta of Women."	Section 1
Average	Interpretation or closely related facts	What specific international convention does the State commit to keeping in line with when eliminating discrimination against women under Section 2?	The Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW).	Section 2
Difficult	Multiple evidence passages, cross-section reasoning, or entity disambiguation	Which government entities are mandated to submit regular reports to Congress on the progress and implementation of this Act, and what must the second report specifically include?	The Philippine Commission on Women (PCW), in coordination with other state agencies and the Commission on Human Rights (CHR). The second report must include an effectiveness assessment and recommendations for amendments.	Section 40

For each question, record:

- The question and its difficulty.
- The expected answer.
- The source document and page, section, or passage containing the evidence.
- All relevant passages if the answer requires multiple pieces of evidence.

Important: All questions must be answerable from your dataset. Difficulty should come from the information needed to answer it, not from vague wording or missing evidence.

3. Testing the Language Model

The existing notebook uses google/flan-t5-base with a baseline chunk size of 1,000 characters and a 200-character overlap.

Your Task:

1. Update the code to use your group's assigned language model.
2. Adjust the prompt and implement necessary guardrails to ensure the model generates accurate answers.

Note: If your assigned model is unable to produce correct answers, you are free to evaluate alternative models.

4. Submit results

Indicate below the final language model and prompt that worked on your dataset and submit this file along with your commented notebook:

- Final Language Model: openai/gpt-oss-20b
- Final Prompt: You are a specialized legal information assistant for Republic Act No. 9710 (The Magna Carta of Women). Answer the user's question clearly and accurately using ONLY the provided document context. If the answer is present in the context, state it directly. If the evidence is insufficient, state exactly: 'I cannot find the answer in the provided document.'